OAgents: A Pre-Standardization Draft Profile for Operational AI Agent Trustworthiness

A Community-Authored Pre-Standardization Draft

JD Longmire
Northrop Grumman Fellow (unaffiliated research)
ORCID: 0009-0009-1383-7698
jdlongmire@outlook.com

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Submission Note

This document is a community-authored pre-standardization draft proposing a behavioral control and evidence profile for operational AI agents. It is not a NIST-issued publication, a consensus standard, or a formally adopted federal requirement. It is intended to inform AI agent standards development, implementation profiling, and public review, and to support routing into future NIST, NCCoE, IEEE, OASIS, IETF, or other voluntary consensus processes.

Requested Reviewer Action

The author requests feedback on four questions:

- Whether this document is appropriately structured as an AI RMF implementation profile for operational AI agents.
- Whether the behavioral-envelope taxonomy is technically coherent and materially distinct from existing identity, authorization, and agent-interoperability efforts.
- Which elements are best advanced through NIST or NCCoE guidance and which belong in voluntary consensus venues such as IEEE, OASIS, or IETF-adjacent workstreams.
- Which control definitions, conformance artifacts, or mappings should be revised before broader circulation.

Abstract

Large language models are increasingly deployed as operational agents that manage infrastructure, generate enterprise content, and execute multi-step workflows. Capability has advanced faster than trust mechanisms. Current agent frameworks support tool use, orchestration, and function execution, but they do not provide standardized guarantees for operational reliability, output quality, or behavioral consistency.

This document proposes OAgents, an AI RMF implementation profile and candidate conformance framework for operational AI agent trustworthiness. The profile defines a behavioral envelope: a structured set of pre-execution gates, post-execution verification controls, and operational-discipline mechanisms. It specifies 26 controls across 7 categories, 3 conformance levels, and observable evidence criteria. The profile is model-agnostic. The behavioral guarantees are properties of the envelope, not of the underlying model.

The OAgents profile maps its control taxonomy to AI RMF functions, categories, and subcategories. A reference implementation demonstrates architectural feasibility for all 26 controls in a development-laboratory environment with live operational use. No generalized empirical benchmarks are claimed. Broader validation across diverse deployment contexts remains community work.

The central claim is practical. Enterprise adoption of operational AI agents requires a portable trust layer above model execution. OAgents proposes that layer as a structured, implementable pre-standardization draft, offered for community review and intended to inform future AI agent standards development.

Foreword

This document is a pre-standardization draft structured as an AI RMF implementation profile, as described in Section 6 of NIST AI 100-1. It is offered for technical review, comparison, and contribution to future standards development. It is not a finalized or adopted profile. Profiles apply AI RMF functions, categories, and subcategories to a defined deployment setting based on the implementing organization's requirements, risk tolerance, and resources.

The OAgents profile addresses a specific use case: AI agents operating in enterprise roles with authority to take consequential actions such as deploying code, managing infrastructure, generating customer-facing content, and responding to incidents. This is a high-autonomy, high-consequence environment. To the author's knowledge, no prior AI RMF implementation profile has addressed this setting at the operational-control level. The author welcomes correction, comparison, and competing approaches.

This draft is offered as a community contribution aligned with NIST's AI Agent Standards Initiative and related NIST and NCCoE work on agent security, identity, authorization, and interoperability. Its purpose is narrower than a general AI governance framework: to propose a concrete trust architecture for operational agents as input to future standards work.

The profile builds on NIST AI RMF 1.0 (NIST AI 100-1) and the Generative AI Profile (NIST AI 600-1). It explicitly addresses the AI 600-1 confabulation risk category, commonly described as hallucination, through the Anti-Hallucination control set. Full mapping to all 12 GAI risk categories is left for a subsequent revision.

Terminology follows AI RMF 1.0 definitions. RFC 2119 normative language, MUST, SHOULD, and MAY, is used in Section 5 to specify conformance requirements.

Development methodology. This standard and its reference implementation were developed using a Human-Curated, AI-Enabled (HCAE) methodology. The human architect defined operational standards, curated behavioral corrections, and determined conformance expectations; the AI operator implemented and maintained the behavioral envelope under those standards. The standard is a direct artifact of operating under its own principles.

1. Introduction

1.1 The Trust Gap in Enterprise AI Agent Operations

Enterprise organizations are deploying AI agents in progressively higher-stakes operational roles: writing and deploying code, managing cloud infrastructure, generating customer-facing documents, and responding to operational incidents. Model capability is improving. The barrier is trust.

When a human operator deploys code to production, the organization trusts that operator because of training, experience, accountability, and process enforcement (code review, CI/CD gates, change management). When an AI agent performs the same action, none of these trust mechanisms exist by default. The agent has no persistent memory of past mistakes. No independent reviewer checks its work. No process gate prevents it from skipping quality checks under prompt pressure. No institutional knowledge carries from one session to the next.

This gap between agent capability and organizational trust is the central barrier to enterprise adoption. The 2025 AI Agent Index underscores the maturity problem: only 4 agents provide agent-specific system cards, 25 of 30 disclose no internal safety results, and 23 of 30 disclose no third-party testing (Casper et al., 2026). This is the risk class the AI RMF is designed to address. The issue is not raw capability. The issue is trustworthy operation.

The supporting standards landscape is still immature. Identity delegation has OAuth and OpenID Connect. Network trust has NIST SP 800-207. Software supply-chain integrity has SLSA and SBOM frameworks. Operational AI agent trust has no widely adopted equivalent. Current NIST and NCCoE work on agent security, identity, authorization, interoperability, and evaluation makes the gap more visible, but control-level guidance for trustworthy agent behavior is still emerging. OAgents is proposed as one contribution to that landscape.

1.2 The OAuth Parallel

In the early 2000s, web applications faced an analogous trust problem. Users needed to grant third-party applications access to their data on other platforms, but the only mechanism was sharing passwords: unverifiable, unscoped, and irrevocable. OAuth solved this by introducing a standardized trust layer: tokens that carry scoped permissions, issued by identity providers, verifiable by any relying party.

OAuth did not replace applications. It added a trust layer that made delegation safer and more portable.

AI agents face the same structural problem. Organizations need to delegate operational authority to AI agents, but the only mechanism is prompt instructions: unverifiable, unenforceable, and forgotten when context compresses. A standardized trust layer for AI agent operations would make delegation safe regardless of which model powers the agent.

The analogy has limits. OAuth addressed a tightly bounded authorization problem with a defined protocol exchange. AI agent trust is broader: it includes context management, output verification,

incident governance, cross-session memory, and organizational discipline. OAgents does not claim OAuth's protocol precision. It claims a similar architectural role: a trust layer above execution that makes delegated operation more governable.

That mechanism is the OAgent behavioral envelope.

1.3 Scope

This profile applies to AI agents operating in the following deployment context:

- Role: Operational agent with authority to take consequential actions (code deployment, infrastructure management, content generation, incident response)
- Autonomy level: Semi-autonomous to autonomous, with defined human oversight thresholds
- Environment: Enterprise operational environments with multiple interdependent systems
- Model: Any large language model (the profile is model-agnostic by design)

This profile does not address pure inference systems without operational authority, consumer-facing conversational AI without consequential action capabilities, or embedded AI in physical systems (robotics, autonomous vehicles).

This profile is scoped to operational reliability and behavioral trust. The AI RMF trustworthiness characteristics of Privacy-Enhanced and Fair with Harmful Bias Managed are out of scope and are not addressed by the behavioral envelope. Organizations deploying OAgents in contexts where these characteristics are material should pair this profile with companion frameworks identified in Appendix B.

1.4 Terminology

OAgent. An AI agent operating within a behavioral envelope that provides structured guarantees about its operational behavior. An OAgent is conformant with this profile when it implements the components specified at the applicable compliance level.

OAgents. The open profile and candidate conformance framework defining behavioral envelope requirements, component taxonomy, and compliance levels for trustworthy AI agent operations.

Behavioral envelope. The structured set of pre-execution gates, post-execution verification, and operational discipline mechanisms that bound an OAgent's behavior. Corresponds to the AI RMF concept of trustworthiness characteristics instantiated as operational controls.

Model-agnostic. Behavioral guarantees that function identically regardless of which large language model provides the underlying computation. The envelope is independent of the model provider.

AI actor (AI RMF definition). Organizations and individuals that play an active role in the AI system lifecycle, including those that deploy or operate AI systems (OECD, 2019, as adopted in NIST AI 100-1).

TEVV. Test, Evaluation, Verification, and Validation, as defined in AI RMF 1.0.

RFC 2119 normative language. MUST indicates a requirement; SHOULD indicates a recommendation; MAY indicates an option.

1.5 Intended Adopters

Infrastructure and platform engineering teams deploying AI agents to manage software systems, cloud infrastructure, CI/CD pipelines, or operational monitoring. These teams need behavioral guarantees equivalent to what they already require of human operators: code review, change management, incident tracking, and documented procedures enforced at the tooling level.

Internal AI platform groups building shared AI agent services for their organizations, accountable for the reliability and trustworthiness of AI capabilities consumed by other teams.

Vendors building AI agent platforms who need to demonstrate behavioral trustworthiness to enterprise customers. OAgents provides a NIST-aligned vocabulary and conformance structure for communicating what behavioral guarantees a platform provides, at what compliance level, and what evidence supports the claim.

This profile is not designed for consumer-facing conversational AI without operational authority, pure inference pipelines, HR or credit decisioning systems, or physical systems where safety-critical real-time constraints require separate certification frameworks.

2. AI RMF Alignment Statement

GOVERN. OAgents behavioral envelope components instantiate governance structures as executable mechanisms: session protocols enforce operational discipline, impact level classification enforces risk-proportionate decision-making, and asset registries maintain organizational accountability.

MAP. OAgents knowledge injection and context classification components operationalize the MAP function's requirement to establish and understand context before acting.

MEASURE. OAgents quality gate components instantiate the MEASURE function's requirement for independent evaluation of AI system outputs. Independent output review, schema validation, security auditing, and hallucination tracking provide the TEVV infrastructure the AI RMF requires.

MANAGE. OAgents enforcement mechanisms, incident lifecycle tracking, and lessons-learned pipeline operationalize the MANAGE function's requirement for active risk response and continuous improvement.

The complete subcategory mapping is provided in Appendix A.

3. Design Principles

3.1 Behavioral Guarantees Must Be Portable Across Model Providers (AI RMF: Valid and Reliable; Secure and Resilient)

Behavioral guarantees must be independent of the underlying model. An organization that encodes operational trust into model-specific prompts or vendor-specific tool configurations has created lock-in at the trust layer, the most critical layer to keep portable. The OAgents profile requires that all behavioral envelope components function identically across model providers.

This is a claim about trust-layer portability, not model equivalence. The profile does not assert that all models perform identically or that model selection is inconsequential. It asserts that the behavioral envelope (memory systems, enforcement gates, quality review agents, and operational

logging) operates at the infrastructure layer, outside the model's context window and independent of any model-specific API. Swapping the model changes the computation; it does not change the behavioral guarantees.

The SHOULD-level recommendation that the independent reviewer come from a different provider reinforces this principle. One provider should not both produce and evaluate the same consequential output, and one provider failure should not compromise both production and verification.

3.2 Enforcement Over Documentation (AI RMF: Accountable and Transparent)

Documented standards are advisory. Under context pressure, deadline urgency, or prompt complexity, AI agents routinely violate documented procedures. The OAgents profile requires that critical behavioral guarantees be implemented as executable gates: mechanisms that block non-compliant actions rather than merely discouraging them. GOVERN 1.4 calls for risk management processes to be established through transparent policies and controls. In this profile, that means commitments with operational consequence should be enforced by executable gates.

3.3 Independent Verification (AI RMF: Accountable and Transparent; Valid and Reliable)

No system can reliably evaluate its own output. An AI agent that generates code and reviews that code in the same context is structurally incapable of independent quality assurance. The OAgents profile requires that quality verification be performed by a separate process with independent context, ideally a different model instance from a different provider. This operationalizes the AI RMF recommendation that TEVV processes include independent review.

3.4 Persistent Learning (AI RMF: Accountable and Transparent; Safe)

AI agent sessions are ephemeral by default. Without explicit persistence mechanisms, every session starts from zero operational knowledge. The OAgents profile requires that operational lessons, behavioral corrections, and contextual knowledge persist across session boundaries and be loaded into every new session. This operationalizes the AI RMF principle of continuous improvement.

3.5 Self-Improving Reliability (AI RMF: Safe; Secure and Resilient)

A compliant behavioral envelope must become more reliable over time. Operational failures must feed back into the envelope's enforcement mechanisms, creating a closed loop from failure detection to automated prevention. This operationalizes the AI RMF's requirement that risk management processes include mechanisms for continual improvement.

4. Behavioral Envelope Architecture

The behavioral envelope interposes structured gates between the user's request and the agent's action, and between the agent's output and its delivery.

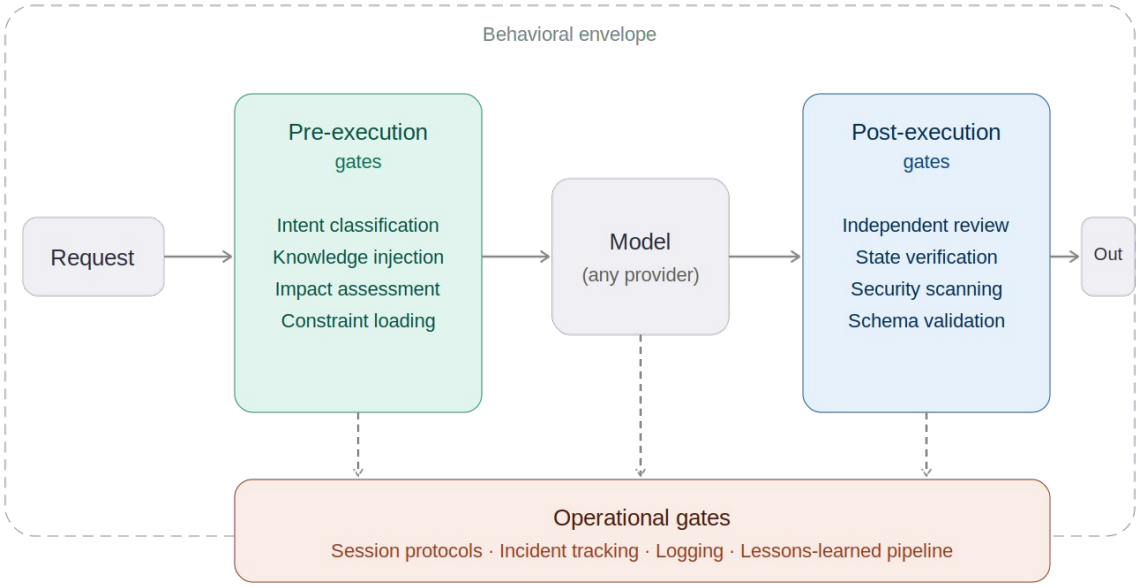

Figure 1. OAgents behavioral envelope architecture.
All agent actions pass through pre- and post-execution gates. Operational gates persist across sessions.

Figure 1. OAgents behavioral envelope architecture. All agent actions pass through pre- and post-execution gates. Operational gates persist across sessions. The model is replaceable; the envelope is not.

4.1 Pre-Execution Gates

Pre-execution gates shape context and constrain behavior before the model processes a request. They implement the AI RMF MAP function's requirement to establish context before acting.

Intent classification. Requests are categorized to determine which tools, constraints, and verification levels apply.

Knowledge injection. Persistent memory from prior sessions (behavioral corrections, project context, operational lessons) is loaded into the model's context.

Impact assessment. Each potential action is classified by blast radius (Level 1: read-only through Level 5: irreversible/external-facing).

Behavioral constraint loading. Domain-specific operational standards, quality requirements, and failure mode catalogs are loaded as active constraints.

4.2 Post-Execution Gates

Post-execution gates verify output quality and safety before delivery. They implement the AI RMF MEASURE function's TEVV requirements.

Independent quality review. A separate model instance reviews significant outputs against defined quality criteria. Results are logged with structured verdicts (pass, pass with warnings, fail).

State verification. Claims about system state are verified against reality using diagnostic commands before being asserted to the user.

Security scanning. Outputs are checked for credential exposure, injection vectors, and information leakage before delivery.

Schema validation. Structured outputs are validated against expected schemas, catching malformed responses before they enter downstream systems.

4.3 Operational Gates

Operational gates impose discipline on the agent's lifecycle across sessions. They implement the AI RMF MANAGE function's requirements for ongoing risk management.

Session protocols. Mandatory startup and shutdown procedures bookend every operational session.

Incident lifecycle management. Operational failures are tracked through a structured lifecycle with cross-session persistence and recurrence detection.

Operational logging. Every significant action is logged as structured data with timestamps and correlation identifiers.

Lessons-learned feedback. Operational failures flow through a multi-stage pipeline: detection, lesson capture, quality criteria update, and enforcement gate installation.

When multiple OAgents operate in coordination, additional principles govern envelope interactions. These coordination principles are specified in Section 12.1.

5. Component Taxonomy

The OAgents profile defines 26 components across 7 categories. Normative language follows RFC 2119: MUST indicates a conformance requirement, SHOULD indicates a recommendation for production deployments, and MAY indicates an optional capability. Each component is mapped to AI RMF 1.0 subcategories in Appendix A.

5.1 Behavioral Shaping

Components that modify the agent's default behavior toward operational reliability. Addresses AI RMF trustworthiness characteristics: Valid and Reliable; Accountable and Transparent.

Component	Level	Description	AI RMF Subcategory
Persistent feedback memory	MUST	Behavioral corrections and confirmations persist across sessions as structured entries with the rule, the rationale, and application guidance. Loaded into every new session context.	GOVERN 1.2; GOVERN 4.1; MANAGE 4.1
Named failure mode catalog	MUST	Known agent failure patterns are explicitly named and described (over-confidence, stale references, hallucinated state, scope creep, instruction drift). The agent self-monitors for cataloged patterns.	MAP 2.2; MEASURE 2.5; MANAGE 1.3

Component	Level	Description	AI RMF Subcategory
Context degradation detection	SHOULD	The agent monitors its own reliability during long sessions, detecting increasing error rates, stale assumptions, or repeated questions. Degradation triggers a recommendation to terminate the session.	MEASURE 2.6; MANAGE 1.1
Reasoning anchors	SHOULD	Stable epistemological commitments that ground reasoning under ambiguity: precision over vagueness, questioning assumptions, verification before assertion.	GOVERN 4.1; MAP 2.3

5.2 Quality Gates

Executable checkpoints that prevent defective work from advancing. Addresses AI RMF trustworthiness characteristics: Valid and Reliable; Accountable and Transparent; Safe.

Component	Level	Description	AI RMF Subcategory
Independent output review	MUST	Significant outputs are reviewed by a separate model instance with independent context. The reviewing model SHOULD be from a different provider than the producing model. Results are logged with structured verdicts.	MEASURE 2.5; MEASURE 2.8; GOVERN 4.3
Process enforcement gate	MUST	Critical workflow steps (e.g., testing before production deployment) are enforced by executable gates that block non-compliant actions. Advisory documentation alone is insufficient for MUST-level guarantees.	GOVERN 1.4; MANAGE 1.2; MANAGE 2.2
Security audit	SHOULD	Automated security checks run against outputs and affected systems before significant actions. Check categories include credential exposure, injection vectors, and access control.	MEASURE 2.7; MANAGE 1.3; GOVERN 6.1
Schema validation	SHOULD	Structured outputs are validated against defined schemas before acceptance into downstream systems, catching malformed outputs from model confabulation.	MEASURE 2.5; MEASURE 1.1

5.3 Operational Discipline

Protocols that impose structure on agent behavior across sessions. Addresses AI RMF trustworthiness characteristics: Accountable and Transparent; Safe; Secure and Resilient.

Component	Level	Description	AI RMF Subcategory
Session lifecycle protocol	MUST	Mandatory start and wrap procedures that verify system state, load context, and ensure clean handoffs. Protocol completion is verified programmatically, not assumed.	GOVERN 1.5; MANAGE 4.1; GOVERN 2.1
Impact level classification	MUST	Actions are classified by blast radius (Level 1: read-only through Level 5: irreversible/external-facing). Higher levels require proportionally more verification and may require human approval.	MAP 3.5; MANAGE 1.1; GOVERN 3.2
Incident lifecycle tracking	SHOULD	Operational failures are tracked through a persistent lifecycle with cross-session continuity, recurrence detection, and severity escalation. Services with 3+ incidents in 30 days are flagged as fragile.	MANAGE 3.2; MEASURE 3.1; MANAGE 2.4
Structured operational logging	SHOULD	Every significant action is logged as structured data with timestamps and correlation identifiers. Logs are append-only and support cross-session pattern analysis.	GOVERN 1.4; MEASURE 3.1; MANAGE 4.2

5.4 Knowledge Injection

Systems that load operational context into the agent's working memory. Addresses AI RMF trustworthiness characteristics: Valid and Reliable; Explainable and Interpretable.

Component	Level	Description	AI RMF Subcategory
Persistent memory system	MUST	Typed memory entries (behavioral feedback, project state, operational context) persist across sessions. Entries include staleness metadata and MUST be verified against current state before being acted upon.	GOVERN 1.2; MAP 1.1; MANAGE 4.1
Domain skill loading	MUST	Operational knowledge (procedures, decision frameworks, service-specific details) is loaded as active context for every session, ensuring consistent standards across sessions.	GOVERN 1.2; MAP 1.1; MAP 3.4
Lessons-learned pipeline	SHOULD	Operational failures flow through a closed-loop pipeline: detection, memory capture, quality criteria update, and enforcement installation. Each stage feeds the next.	MANAGE 4.2; GOVERN 5.2; MEASURE 4.1

5.5 Enforcement Mechanisms

Components that make behavioral guarantees executable rather than advisory. Addresses AI RMF trustworthiness characteristics: Accountable and Transparent; Safe.

Component	Level	Description	AI RMF Subcategory
Executable action gates	MUST	Critical behavioral guarantees are implemented as gates that block non-compliant actions at the tooling level (e.g., version control hooks, CI/CD checks). Prompt-level instructions alone are insufficient for MUST-level guarantees.	GOVERN 1.4; MANAGE 2.2; GOVERN 4.3
Severity escalation rules	SHOULD	Recurrence of operational failures automatically escalates severity. First occurrence generates a warning; second generates an error; third generates a critical alert and structural review.	MANAGE 1.3; MANAGE 3.2; MEASURE 3.2
Protocol compliance verification	SHOULD	Session protocol completion is verified programmatically at session wrap. Non-completion blocks session closure until resolved.	GOVERN 1.5; MANAGE 2.2; GOVERN 2.1

5.6 Project Governance

Components that impose organizational discipline on multi-system operations. Addresses AI RMF trustworthiness characteristics: Accountable and Transparent; Secure and Resilient.

Component	Level	Description	AI RMF Subcategory
Centralized work tracking	SHOULD	A single backlog spanning all managed systems, with structured priority and status fields. Reviewed and updated during every session wrap.	GOVERN 1.6; MANAGE 4.2; GOVERN 2.1
Platform sovereignty	SHOULD	Critical operational data exists on at least two independent platforms simultaneously. No single vendor removal can cause data loss or operational disruption.	GOVERN 6.2; MANAGE 1.4; MAP 4.2
Asset registry	SHOULD	A maintained inventory of all managed systems, accounts, and platforms with sync status and gap identification. Updated when repository-level changes occur.	GOVERN 1.6; MAP 4.1; MAP 4.2
Vendor independence	SHOULD	Product-critical automation lives in version-controlled code, not in third-party workflow tools. No opaque vendor-managed state for critical operations.	GOVERN 6.1; GOVERN 6.2; MAP 4.1

5.7 Anti-Hallucination

Components that detect and prevent false assertions. Addresses AI RMF trustworthiness characteristics: Valid and Reliable; Accountable and Transparent. Addresses NIST AI 600-1 GAI risk: Confabulation (colloquially, hallucination).

Component	Level	Description	AI RMF Subcategory
State verification protocol	MUST	Before asserting any claim about system state, the agent MUST verify against current reality using appropriate diagnostic methods. Memory of a state is not confirmation of a state.	MEASURE 2.5; MAP 2.2; GOVERN 4.2
Memory staleness detection	MUST	Memory entries are treated as claims about past state. Before acting on a memory, the agent verifies it against current system state. Conflicts are resolved in favor of observed reality.	MEASURE 2.6; MAP 1.1; MANAGE 1.3
Hallucination tracking	SHOULD	Detected hallucinations are logged with the false claim, the truth, and the detection method. Recurrence patterns trigger quality criteria updates. Resolution requires 10+ subsequent clean reviews of the same type.	MEASURE 3.1; MEASURE 4.1; MANAGE 4.2

6. Conformance, Evidence, and Certification Path

6.1 How Conformance Is Established

OAgents conformance is established through evidence: observable artifacts demonstrating that the specified components are implemented and operational. Conformance is not established by assertion. A system that claims to implement independent output review without logs showing distinct reviewing model identifiers and structured verdicts is not conformant, regardless of how the system is described.

The complete set of evidence criteria for MUST-level components is specified in Appendix C. Each MUST component requires 2-3 observable artifacts. Evidence must be producible on demand by an internal audit function or external assessor.

6.2 Conformance Verification by Level

Level 1 (OAgent-Basic), Self-Assessment. Conformance is verified by the deploying organization against the Appendix C checklist. Self-assessment is appropriate for initial deployment, internal governance, and low-stakes operational contexts.

Level 2 (OAgent-Standard), Documented Evidence Review. Conformance is verified through a structured evidence review conducted by an internal audit team independent of the deployment team, or by an external reviewer. The review examines actual artifacts (session logs, review records, enforcement gate configurations, memory store contents) against the conformance criteria in Appendix C. Documented evidence review is appropriate for production deployments with periodic human oversight.

Level 3 (OAgent-Autonomous), Third-Party Verification. Conformance is verified by an independent third-party assessor who has no organizational relationship with the deploying organization. The third-party assessor model is analogous to FedRAMP's Third Party Assessment Organization (3PAO) structure.

6.3 Certification Status

Formal certification programs for OAgents conformance are not yet established. This document defines the conformance criteria and evidence requirements that a certification program would use. Level 2 conformance criteria are structured to support future mapping to FedRAMP Moderate-relevant control families for AI agent deployments; a formal alignment mapping is planned as a future deliverable.

7. Compliance Levels

The OAgents profile defines three conformance levels to support incremental adoption consistent with the AI RMF's design as a flexible, voluntary framework.

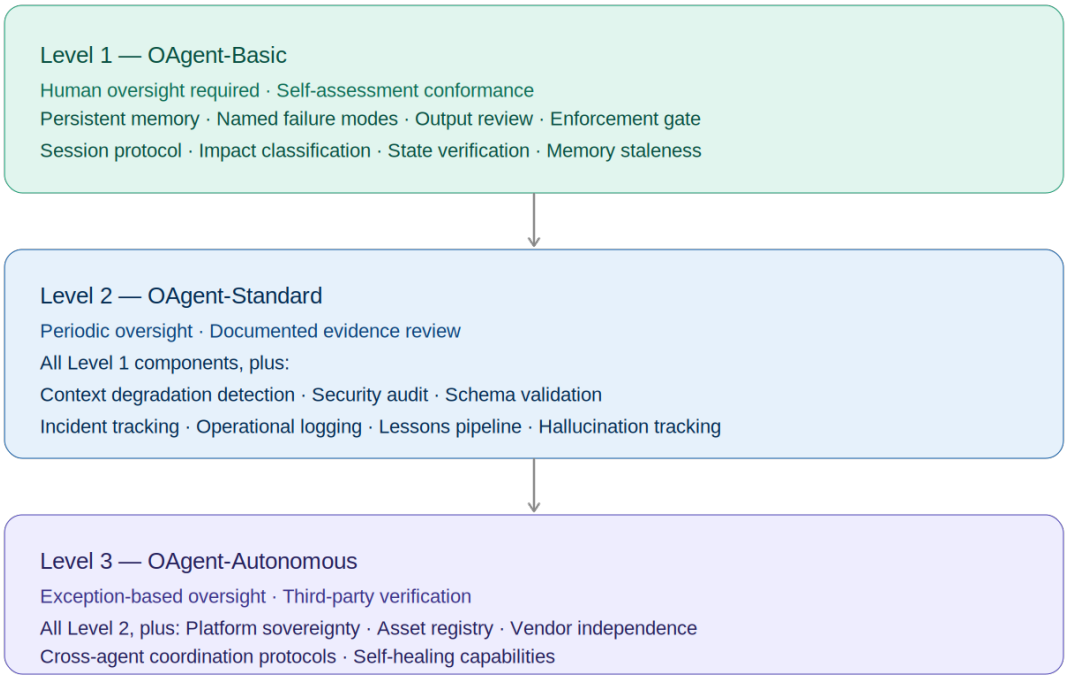

Figure 2. OAgents compliance levels. Each level adds components and increases conformance verification rigor.

Figure 2. OAgents compliance levels. Each level adds components and increases conformance verification rigor.

Level 1: OAgent-Basic

All MUST components are implemented. Suitable for supervised operational use with human oversight. Implements: Persistent feedback memory, Named failure mode catalog, Independent output review, At least one process enforcement gate, Session lifecycle protocol, Impact level classification, Persistent memory system with staleness detection, Domain skill loading, At least one executable action gate, State verification before all assertions, Memory staleness detection.

Trustworthiness assurance: Level 1 provides baseline guarantees of valid and reliable behavior, minimal accountability structures, and anti-hallucination protections.

Level 2: OAgent-Standard

All MUST and SHOULD components are implemented. Suitable for production use with periodic human oversight. Adds to Level 1: Context degradation detection, Reasoning anchors, Security auditing, Schema validation, Incident lifecycle tracking, Structured operational logging, Lessons-learned pipeline, Severity escalation rules, Protocol compliance verification, Centralized work tracking, Hallucination tracking.

Trustworthiness assurance: Level 2 provides comprehensive operational guarantees across all seven AI RMF trustworthiness characteristics.

Level 3: OAgent-Autonomous

Level 2 compliance plus: Platform sovereignty (multi-platform data replication), Asset registry maintenance, Vendor independence verification, Cross-agent coordination protocols consistent with Section 12.1, and self-healing capabilities (automatic remediation of known failure patterns without human intervention).

Trustworthiness assurance: Level 3 provides guarantees suitable for high-autonomy deployments with minimal human oversight.

8. Compounding Effects

Individual components provide linear value. The OAgents profile architecture creates five compounding effects in which component interactions produce multiplicative reliability gains over time.

8.1 Lesson-to-Enforcement Pipeline (AI RMF: MANAGE 4.2; GOVERN 5.2)

Failure detection (operational logging) feeds lesson capture (persistent memory), which updates quality criteria (independent review), which installs enforcement gates (executable action gates). Four components create a closed loop where every failure permanently strengthens the envelope.

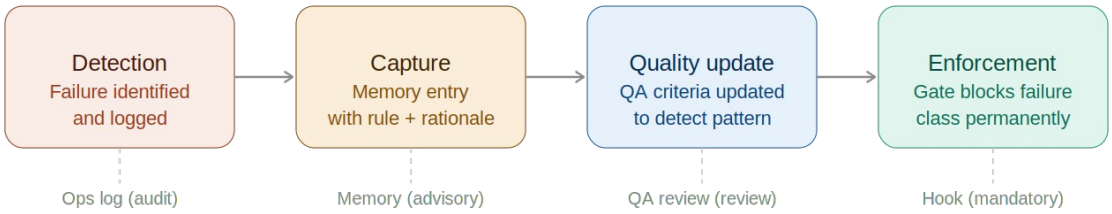

Figure 3. Lesson-to-enforcement pipeline. Every operational failure produces four layers of defense against recurrence. The system strengthens from its failures rather than repeating them.

Figure 3. Lesson-to-enforcement pipeline. Every operational failure produces four layers of defense against recurrence. The system strengthens from its failures rather than repeating them.

8.2 Advisory Plus Mandatory (GOVERN 1.4; MANAGE 2.2). Behavioral memory shapes agent behavior (advisory). Executable gates block violations (mandatory). Advisory systems degrade

under context pressure. Mandatory systems do not. The combination is strictly stronger than either mechanism alone.

8.3 Cross-Session Continuity (MANAGE 4.1; GOVERN 1.5). Persistent memory, session protocols, and operational logging together create continuous operational awareness across inherently ephemeral sessions.

8.4 Gap-to-Governance (GOVERN 5.2; MANAGE 4.2). When operational gaps are discovered during agent execution, the lessons-learned pipeline installs structural fixes within the same session.

8.5 Sovereignty by Default (GOVERN 6.1; GOVERN 6.2). Platform sovereignty and vendor independence together create a system that is inherently portable. No single vendor dependency exists for critical operational state.

9. Reference Implementation

The OAgents behavioral envelope has been implemented as a reference system operating in a development laboratory context. The reference implementation runs on a multi-host Linux environment and manages a suite of containerized services including authentication, collaboration, communication, and version control systems. This is a single-organization development environment, not a scaled production deployment; it is used for active development and operational management of the Ologos Corp technical infrastructure. The AI operator is a large language model accessed via API, with a separate model instance from a different provider performing independent quality review.

A sanitized snapshot of the reference implementation, with credentials, internal addresses, and identifying information removed but with all structural artifacts, script logic, hook mechanisms, memory schemas, and log formats intact, is published at https://github.com/ologos-corp/OAgents-standard/tree/main/reference. This snapshot constitutes the evidence artifacts that Appendix C requires conforming implementations to produce.

Human-Curated, AI-Enabled (HCAE) methodology. The reference implementation was developed and operated using an HCAE methodology: a human architect defined operational standards, curated behavioral corrections, and determined conformance expectations; an AI operator implemented and maintained the behavioral envelope under those standards. This is not incidental to the profile's design; it is the operational model the profile is built to support and constrain.

9.1 Reference Stack Summary

Component Category	Implementation Approach	Primary Artifact	Reference File
Behavioral Shaping	YAML-frontmatter memory entries with typed fields (rule, rationale, application guidance, date)	Structured memory store + session startup load log	memory/ example_feedback.md, scripts/session-start.py

Component Category	Implementation Approach	Primary Artifact	Reference File
Quality Gates	Separate model instance (different provider) called via API; structured JSON verdict (pass/warnings/fail) logged per review	QA review log with model IDs and verdicts	qa-agent/agent.py, ops/examples/qa_log.json
Operational Discipline	Python session scripts enforcing mandatory start/wrap procedures; wrap script verifies git clean, QA confirmed, logs pushed	Session startup and wrap logs with timestamps and compliance status	scripts/session-start.py, scripts/session-wrap.py
Knowledge Injection	Markdown skill files and memory index loaded into model context at session start; staleness metadata in memory frontmatter	Skill files + memory index with load confirmation in session log	skills/sysadmin.md, memory/MEMORY.md
Enforcement Mechanisms	Git pre-commit hook blocks commit without QA review completion; pre-push hook enforces dev-first workflow and security scan	Hook source code (version-controlled) + activation log	hooks/pre-commit, hooks/pre-push
Project Governance	Structured JSON ops logs (append-only); centralized backlog tracked in GitHub Projects; asset registry in ops directory	Ops log files with schema definitions	ops/README.md, ops/examples/sysadmin_log.json
Anti-Hallucination	State verification commands executed before assertions; memory staleness check on load; confabulation entries logged to QA log	Verification commands in session scripts; QA log entries flagging false assertions	skills/quality_assurance.md, qa-agent/prompts/qa_review.md

9.2 Operational Metrics

The following metrics are drawn from live operational logs of the reference implementation. The reference implementation operates as a development laboratory environment: a single-organization system used for active development and operational management, not a scaled

production deployment. Structured logging has been active since 2026-04-02; the underlying system has been in continuous operation since approximately mid-March 2026.

Metric	Count	Source
Git commits (each triggers pre-commit and pre-push hook enforcement)	367	git log
Independent QA reviews executed	83	ops/qa_log.json
QA pass rate	97.6% (81 pass, 2 fail)	ops/qa_log.json
Automated health check executions	349	ops/ sysadmin_log.json
Security audit runs	4	ops/security_log.json
Lessons learned captured (lesson-to-enforcement pipeline)	9	ops/ lessons_learned.json
Behavioral correction memories accumulated	23	memory/ feedback_*.md
Total persistent memory entries	49	memory/*.md
Confabulation catches logged	2	ops/qa_log.json

These figures demonstrate operational use, not empirical benchmarking. The 367 commits with associated hook enforcement events show the enforcement gate has been active across the full project lifetime. The 83 independent QA reviews, each performed by a model instance from a different provider than the producing agent, show the independent review mechanism has been exercised at meaningful volume. No repeat occurrence of the same failure pattern has been observed after enforcement installation.

9.3 Scope and Limitations

The reference implementation demonstrates the architectural feasibility of all 26 components. It is not offered as empirical validation of the profile's effectiveness claims. Systematic empirical validation is identified as community work. The conformance evidence criteria in Appendix C define what that measurement should look like. The community is invited to conduct that measurement using the reference implementation as a starting point and to contribute results for incorporation in subsequent revisions.

The appropriate sequence for a v1.0 specification is: define the taxonomy, demonstrate feasibility, invite community validation. That is the sequence this document follows.

10. Related Work

Agent Frameworks (LangChain, CrewAI, AutoGen, MetaGPT). These frameworks address agent orchestration. Behavioral trustworthiness during execution is outside their scope. OAgents addresses the trust layer that sits above any agent framework.

Workflow Automation (n8n, Zapier, Apache Airflow). These platforms provide workflow orchestration for structured, repeatable processes. OAgents addresses the behavioral governance of judgment-exercising agents, a problem that workflow orchestration does not encounter.

AI Development Environments (GitHub Copilot, Cursor, Claude Code). These tools are designed for interactive developer assistance. OAgents addresses autonomous or semi-autonomous operational agents with persistent memory, cross-session accountability, and enforcement-backed behavioral constraints.

Output Validation Frameworks (Guardrails AI, NeMo Guardrails). These frameworks correspond to the post-execution gate layer of the OAgents behavioral envelope. OAgents encompasses output validation within a broader framework that additionally addresses pre-execution shaping, cross-session memory, operational discipline, and governance.

Agent Behavioral Contracts (ABC, Bhardwaj 2026). ABC focuses on formal runtime specification and enforcement within a session. OAgents focuses on operational completeness across sessions. The frameworks could be composed: ABC governing runtime behavioral invariants, OAgents governing the cross-session operational envelope.

ISO/IEC 42001. Operates at the organizational management system level. OAgents operates at the individual agent level. These are complementary scopes.

NIST AI RMF 1.0 (NIST AI 100-1, 2023). The foundational governance framework this profile extends. OAgents is not a competitor to the AI RMF; it is an implementation profile of it, for the specific context of enterprise AI agent operations.

NIST AI 600-1, Generative AI Profile (2024). OAgents explicitly addresses confabulation through the anti-hallucination component category. Full mapping to all 12 GAI risk categories is planned for a subsequent revision.

NIST CAISI and NCCoE AI agent activities (2026). NIST's AI Agent Standards Initiative and the NCCoE concept paper on software and AI agent identity and authorization highlight an active federal interest in agent security, interoperability, authorization, auditing, and non-repudiation. OAgents is designed to be consistent with and supportive of that emerging landscape.

NIST SP 800-207, Zero Trust Architecture (2020). SP 800-207 provides the conceptual precedent: trust must be verified continuously through explicit mechanisms, not assumed based on network location or model capability. The behavioral envelope applies Zero Trust principles to AI agent operations.

11. Adoption Path

Phase 1 -- Specification (Current). Publish the component taxonomy, conformance levels, and AI RMF mapping as an open specification. Published via Zenodo (DOI: https://doi.org/10.5281/zenodo.19425021). Standards track: Share with NIST AI RMF community as an implementation profile example and with CAISI and NCCoE as a candidate technical framework for operational agent trust.

Phase 2 -- Reference Implementations. Open-source reference implementations of the behavioral envelope for popular agent frameworks (LangChain, CrewAI, raw API). Standards track: Public comment period on v1.0 specification. Publish v1.1 with revised component definitions and expanded SHOULD-level conformance evidence criteria.

Phase 3 -- Tooling and Federal Alignment. Development tools for building OAgent-conformant systems: compliance checkers, behavioral envelope scaffolding, and testing frameworks. Federal alignment: Level 2 (OAgent-Standard) is intended to support future mapping to FedRAMP Moderate-relevant control families for AI agent deployments. A formal alignment mapping is planned as a Phase 3 deliverable, in coordination with GSA and relevant Agency AOs. Standards track: Engage IEEE, OASIS, and where appropriate IETF-adjacent communities.

Phase 4 -- Ecosystem and Certification. Independent quality review services. Pre-built domain-specific behavioral memory sets. A certification program modeled on FedRAMP's 3PAO structure. Standards lifecycle: v1.0 (draft specification) → v1.1 (community comment incorporation) → v2.0 (multi-agent coordination formalized, NIST AI 600-1 full coverage, FedRAMP alignment published).

12. Limitations and Open Problems

Context window cost. The behavioral envelope consumes model context capacity. Risk-proportional context loading is the recommended mitigation. Optimal context budget management strategies are an open research question.

Bootstrap period. A new OAgent deployment starts with an empty behavioral envelope. Pre-built behavioral templates by domain can reduce bootstrap time; developing such templates is identified as community work.

Verification latency. Independent quality review and state verification add latency to every operation. Risk-proportional verification is the recommended mitigation, but requires accurate impact classification.

Empirical validation. Systematic empirical measurement of the profile's effectiveness remains to be conducted across diverse deployment contexts by the community. Appendix C defines what that measurement should look like.

Multi-agent coordination. The current specification is fully defined for single-agent deployments. Coordination principles for multi-agent environments are described in Section 12.1; full formal specification is future work.

Formal verification. Mathematical proof that enforcement gates cannot be bypassed, that memory staleness detection is complete, that the lesson-to-enforcement pipeline terminates, is identified as future work.

NIST AI 600-1 full coverage. The current profile explicitly addresses the confabulation risk category. Full mapping to all 12 GAI risk categories is planned for a subsequent revision.

12.1 OAgents in Multi-Agent Environments

Many high-value operational deployments involve multiple AI agents coordinating on shared tasks. This specification addresses the behavioral envelope of each individual agent. The following principles govern OAgents coordination in multi-agent environments, pending full formalization in a subsequent revision.

Each OAgent maintains its own envelope. A compliant OAgent does not inherit behavioral guarantees from the orchestrating agent. An orchestrated OAgent MUST apply its own pre-execution gates to instructions received from another agent.

Incident ownership follows action authority. The agent that takes a consequential action is the incident owner for that action. The orchestrator SHOULD maintain a cross-agent incident summary for pipeline-level observability.

Independent quality review applies at output boundaries. A compliant implementation SHOULD include at least one independent review gate at each significant inter-agent handoff, not only at the final pipeline output.

Shared memory requires versioned access. Memory entries MUST include an authoring agent identifier and a timestamp. Agents MUST NOT overwrite memory entries authored by other agents without explicit conflict resolution.

Enforcement gates are non-negotiable across agents. An orchestrating agent cannot instruct a subordinate OAgent to bypass its enforcement gates. An OAgent that receives an instruction to skip a MUST-level component MUST treat this as a violation of its behavioral contract and either refuse the instruction or escalate to human oversight.

13. Conclusion

Enterprise AI agents operating in high-autonomy roles such as code deployment, infrastructure management, and incident response face a trust gap that model capability alone cannot close. The standards infrastructure to address this gap is nascent: identity delegation has OAuth, network trust has SP 800-207, but a control and conformance profile for operational AI agent behavioral trust has no widely adopted equivalent. OAgents contributes to that developing landscape through a behavioral-envelope profile that makes AI agent operations reliable, auditable, and self-improving for enterprise operational contexts, regardless of which model powers the agent.

This draft is structured as an AI RMF Implementation Profile, grounding its 26 components in the NIST AI RMF 1.0 functions, categories, and subcategories, and offered as pre-standardization input for community evaluation and future consensus development. The taxonomy is open. No organization needs permission to implement the profile. Any organization that implements the required controls at the stated conformance level may claim conformance on the terms defined here.

We invite NIST, NCCoE, IEEE, OASIS, and the broader standards community to evaluate this taxonomy, contribute replication evidence, propose revisions, and engage in public comment and consensus processes as the profile matures. We specifically invite federal agencies, regulated enterprises, and enterprise platform teams to evaluate OAgents as an implementation framework for AI agent trustworthiness, auditability, and operational discipline.

OAgents offers a reviewable, implementable profile for behavioral trust in operational AI agents. Its control taxonomy is explicit, its evidence criteria are observable, and its reference implementation is available for scrutiny and replication.

13A. Suggested Submission Cover Memo

The following text may be used as a concise submission note when sharing this document with NIST, NCCoE, or other standards-facing reviewers.

Subject: Pre-standardization draft profile for trustworthy operational AI agents

Dear Reviewer,

Attached is OAgents: A Pre-Standardization Draft Profile for Operational AI Agent Trustworthiness. The document is a community-authored pre-standardization draft aligned with the NIST AI Risk Management Framework and with current NIST and NCCoE work on AI agent security, identity, authorization, interoperability, and evaluation.

OAgents proposes a behavioral control and evidence profile for operational AI agents that take consequential actions in enterprise environments. The draft defines 26 controls across 7 categories, three conformance levels, and observable evidence criteria intended to support implementation, comparison, and future consensus development.

This submission is not offered as a claim of NIST endorsement or as a request for immediate federal adoption. It is a pre-standardization contribution. The request is narrower: consideration as input to AI agent standards development, feedback on technical fit, and routing guidance regarding the most appropriate pathway or collaborative venue.

Respectfully,

JD Longmire

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Appendix A: AI RMF Subcategory Mapping

This appendix provides the complete mapping between OAgents components and AI RMF 1.0 subcategories. Mappings are drawn from the primary source: NIST AI 100-1 (January 2023), Tables 1-4. The AI RMF subcategory labels are verified primary-source anchors; the component-to-subcategory assignments are authorial interpretive mappings proposed for community scrutiny.

A.1 GOVERN Function Mappings

AI RMF Subcategory	OAgents Component(s)
GOVERN 1.1: Legal and regulatory requirements are understood, managed, and documented.	Platform sovereignty; Vendor independence; Asset registry
GOVERN 1.2: Trustworthy AI characteristics are integrated into organizational policies and practices.	Persistent feedback memory; Persistent memory system; Domain skill loading

AI RMF Subcategory	OAgents Component(s)
GOVERN 1.3: Risk management activities are determined based on risk tolerance.	Impact level classification
GOVERN 1.4: Risk management processes are established through transparent policies and controls.	Process enforcement gate; Executable action gates; Structured operational logging
GOVERN 1.5: Ongoing monitoring and periodic review are planned with defined roles.	Session lifecycle protocol; Protocol compliance verification
GOVERN 1.6: Mechanisms are in place to inventory AI systems.	Asset registry; Centralized work tracking
GOVERN 2.1: Roles and responsibilities for AI risk management are documented and clear.	Session lifecycle protocol; Centralized work tracking; Protocol compliance verification
GOVERN 2.2: Personnel receive AI risk management training.	Domain skill loading; Named failure mode catalog
GOVERN 3.2: Policies define roles for human-AI configurations and oversight.	Impact level classification; Session lifecycle protocol
GOVERN 4.1: Organizational practices foster a critical thinking and safety-first mindset.	Persistent feedback memory; Reasoning anchors; Named failure mode catalog
GOVERN 4.2: Teams document risks and communicate about impacts.	Structured operational logging; Hallucination tracking; State verification protocol
GOVERN 4.3: Practices enable AI testing, incident identification, and information sharing.	Independent output review; Incident lifecycle tracking; Security audit
GOVERN 5.2: Mechanisms enable regular incorporation of adjudicated feedback into design.	Lessons-learned pipeline; Persistent feedback memory
GOVERN 6.1: Policies address AI risks associated with third-party entities.	Vendor independence; Security audit; Asset registry

AI RMF Subcategory	OAgents Component(s)
GOVERN 6.2: Contingency processes handle failures in third-party systems.	Platform sovereignty; Incident lifecycle tracking; Severity escalation rules

A.2 MAP Function Mappings

AI RMF Subcategory	OAgents Component(s)
MAP 1.1: Intended purposes, context, and prospective settings are understood and documented.	Persistent memory system; Domain skill loading; Knowledge injection
MAP 2.2: AI system knowledge limits and human oversight of outputs are documented.	Named failure mode catalog; Context degradation detection; State verification protocol
MAP 2.3: Scientific integrity and TEVV considerations are identified and documented.	Reasoning anchors; Schema validation; Independent output review
MAP 3.4: Processes for operator proficiency with AI system performance are defined and documented.	Domain skill loading; Named failure mode catalog; Session lifecycle protocol
MAP 3.5: Processes for human oversight are defined and documented.	Impact level classification; Session lifecycle protocol
MAP 4.1: Approaches for mapping AI technology and legal risks of components are in place.	Vendor independence; Asset registry; Security audit
MAP 4.2: Internal risk controls for AI system components are identified and documented.	Platform sovereignty; Asset registry; Process enforcement gate

A.3 MEASURE Function Mappings

AI RMF Subcategory	OAgents Component(s)
MEASURE 1.1: Approaches and metrics for measurement of AI risks are selected and implemented.	Independent output review; Schema validation; Hallucination tracking

AI RMF Subcategory	OAgents Component(s)
MEASURE 2.5: AI system effectiveness is evaluated through TEVV processes.	Independent output review; State verification protocol; Schema validation
MEASURE 2.6: AI system performance is monitored for change.	Context degradation detection; Memory staleness detection; Incident lifecycle tracking
MEASURE 2.7: AI system security and resilience are evaluated.	Security audit; Executable action gates; Process enforcement gate
MEASURE 2.8: Risks from AI system use and deployment are evaluated.	Independent output review; Impact level classification; Named failure mode catalog
MEASURE 3.1: Mechanisms for tracking identified AI risks over time are in place.	Structured operational logging; Incident lifecycle tracking; Hallucination tracking
MEASURE 3.2: Risk tracking approaches are evaluated for effectiveness.	Severity escalation rules; Lessons-learned pipeline
MEASURE 4.1: Measurement approaches are identified and evaluated for improvement.	Lessons-learned pipeline; Hallucination tracking; Independent output review

A.4 MANAGE Function Mappings

AI RMF Subcategory	OAgents Component(s)
MANAGE 1.1: Risks are prioritized based on assessment.	Impact level classification; Context degradation detection
MANAGE 1.2: Responses to risks are developed and planned.	Process enforcement gate; Executable action gates; Session lifecycle protocol
MANAGE 1.3: Risk response plans are implemented and documented.	Named failure mode catalog; Severity escalation rules; Memory staleness detection
MANAGE 1.4: Responses to third-party risks are planned.	Platform sovereignty; Vendor independence; Asset registry
MANAGE 2.2: Mechanisms are in place to respond to emergent risks.	Executable action gates; Process enforcement gate; Severity escalation rules

AI RMF Subcategory	OAgents Component(s)
MANAGE 2.4: Identified risks are monitored and resources allocated.	Incident lifecycle tracking; Centralized work tracking; Structured operational logging
MANAGE 3.2: Incidents are monitored, documented, and reported.	Incident lifecycle tracking; Structured operational logging; Severity escalation rules
MANAGE 4.1: Residual risks are documented and managed.	Persistent feedback memory; Persistent memory system; Session lifecycle protocol
MANAGE 4.2: Processes are in place for continual improvement.	Lessons-learned pipeline; Structured operational logging; Hallucination tracking

Appendix B: Trustworthiness Characteristic Coverage

This appendix maps OAgents compliance levels to the AI RMF 1.0 trustworthiness characteristics (NIST AI 100-1, Section 3).

Trustworthiness Characteristic	Level 1	Level 2	Level 3
Valid and Reliable	Partial: state verification, output review	Full: adds schema validation, degradation detection	Full
Safe	Partial: enforcement gates, impact classification	Full: adds security audit, incident tracking	Full
Secure and Resilient	Partial: executable gates	Substantial: adds security audit, logging	Full: adds sovereignty, vendor independence
Accountable and Transparent	Partial: session protocols, output review logging	Full: adds structured logging, compliance verification	Full
Explainable and Interpretable	Partial: named failure modes, reasoning anchors	Substantial: adds hallucination tracking	Full
Privacy-Enhanced	Not directly addressed -- implementer responsibility	Not directly addressed -- implementer responsibility	Not directly addressed

Trustworthiness Characteristic	Level 1	Level 2	Level 3
Fair with Harmful Bias Managed	Not directly addressed -- implementer responsibility	Not directly addressed -- implementer responsibility	Not directly addressed

Privacy-Enhanced and Fair with Harmful Bias Managed are not addressed by the OAgents behavioral envelope, which is scoped to operational reliability for enterprise AI agents managing software and infrastructure. This is a deliberate scope decision, not an oversight. Organizations deploying OAgents in contexts where these characteristics are material should pair this profile with: NIST Privacy Framework 1.0; NIST SP 1270 for bias identification and mitigation; NIST AI 600-1 (GenAI Profile) Sections on Harmful Bias and Homogenization; and ISO/IEC 23894:2023 for integrated AI risk management.

Appendix C: Conformance Evidence

This appendix defines observable criteria for each MUST-level component. Conformance is demonstrated by producing the specified evidence artifacts, which may be reviewed by an internal audit function or external assessor. Evidence requirements are proportionate to the component's risk significance. Evidence completeness (the ability to produce all specified artifacts on demand) is the conformance requirement; it is not expected that all artifacts will exist from day one of a deployment. Implementations build toward complete evidence as the system matures.

C.1 Behavioral Shaping

Persistent feedback memory (MUST)

- Evidence 1: A structured memory store containing at minimum: rule text, rationale, date captured, and source session identifier for each entry.
- Evidence 2: Session startup logs showing memory entries were loaded at the beginning of each session, with entry count and load timestamp.
- Evidence 3: Demonstration that at least one behavioral correction entry exists and was applied in a subsequent session.

Named failure mode catalog (MUST)

- Evidence 1: A documented catalog of named failure modes applicable to the agent's operational context, with description and detection indicators for each.
- Evidence 2: Session logs or review verdicts referencing at least one failure mode by name during the evaluation period.

C.2 Quality Gates

Independent output review (MUST)

- Evidence 1: Review logs showing for each reviewed output: timestamp, producing model identifier, reviewing model identifier (must differ), verdict, and specific findings where verdict is not clean pass.

- Evidence 2: Confirmation that reviewing model is from a different provider, or documented rationale where provider diversity is not feasible.
- Evidence 3: Aggregate pass/fail statistics covering the evaluation period, with total review count as denominator.

Process enforcement gate (MUST)

- Evidence 1: At least one executable gate implemented at the tooling level that blocks a defined class of non-compliant action.
- Evidence 2: Logs or test results demonstrating the gate fires on a known-non-compliant input and blocks the action.
- Evidence 3: Documentation of which process invariants the gate enforces.

C.3 Operational Discipline

Session lifecycle protocol (MUST)

- Evidence 1: Session startup logs showing system health check, pending incident review, and context loading were performed, with timestamps.
- Evidence 2: Session wrap logs showing state verification was performed, with pass/fail status for each check.
- Evidence 3: Protocol compliance rate over the evaluation period (target: 100%).

Impact level classification (MUST)

- Evidence 1: A defined impact level scale (minimum: 3 levels distinguishing read-only, reversible write, and irreversible/external-facing actions).
- Evidence 2: Action logs showing impact level was assessed and recorded for consequential actions during the evaluation period.
- Evidence 3: At least one instance demonstrating elevated verification or human approval was triggered by a high-impact classification.

C.4 Knowledge Injection

Persistent memory system (MUST)

- Evidence 1: Memory store structure showing typed entries with staleness metadata (last-verified timestamp or equivalent).
- Evidence 2: Session logs showing at least one instance of staleness detection: a memory entry verified against current system state before being acted upon.
- Evidence 3: Conflict resolution log or policy showing how memory entries that conflict with observed reality are handled.

Domain skill loading (MUST)

- Evidence 1: At least one domain skill file loaded into agent context at session start, with load confirmed in session startup log.

- Evidence 2: Demonstration that skill content was applied during a session: an action traceable to a loaded skill's guidance.

C.5 Enforcement Mechanisms

Executable action gates (MUST)

- Evidence 1: At least one action gate implemented at the tooling level, distinct from prompt-level instructions.
- Evidence 2: Gate source code or configuration demonstrating it is version-controlled and not modifiable by the agent at runtime.
- Evidence 3: Test or log showing gate activation on a non-compliant action.

C.6 Anti-Hallucination

State verification protocol (MUST)

- Evidence 1: Session logs showing verification commands were executed before system state claims were made to the user.
- Evidence 2: At least one log entry showing a state claim that was corrected or withheld because verification failed.

Memory staleness detection (MUST)

- Evidence 1: Memory system records showing last-verified timestamps on entries.
- Evidence 2: At least one instance where a memory entry was verified and either confirmed or updated based on current system state, with the verification action logged.

C.7 Conformance Summary Checklist

The following checklist may be used by an internal audit function or external assessor to verify Level 1 (OAgent-Basic) conformance.

Component	Evidence Required	Evidence Present	Notes
Persistent feedback memory	Memory store + load logs + applied entry		
Named failure mode catalog	Catalog document + session references		
Independent output review	Review logs with model IDs + aggregate stats		
Process enforcement gate	Gate implementation + block demonstration		
Session lifecycle protocol	Startup logs + wrap logs + compliance rate		

Component	Evidence Required	Evidence Present	Notes
Impact level classification	Scale definition + action logs + escalation instance		
Persistent memory system	Typed entries + staleness metadata + conflict policy		
Domain skill loading	Skill files + load logs + applied instance		
Executable action gates	Gate source + version control + activation log		
State verification protocol	Verification commands in logs + corrected claim		
Memory staleness detection	Timestamps + verified/updated entry log		

Level 2 (OAgent-Standard) conformance requires the above plus evidence for all SHOULD components. Detailed evidence criteria for SHOULD components will be specified in a subsequent revision of this profile.

Corresponding author: JD Longmire, Northrop Grumman Fellow (unaffiliated research). jdlongmire@outlook.com. ORCID: 0009-0009-1383-7698

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This document is offered as a community implementation profile aligned with the NIST AI RMF. Comments and replication results may be directed to the author.